

SNEHA PANDIT

AI/ML Engineer | Generative AI • NLP • Multimodal AI

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EDUCATION

GLA University | B.Tech in Computer Science & Engineering (Spec. AI & ML) | Expected: 2027

TECHNICAL SKILLS

- AI/ML & GenAI: PyTorch, Hugging Face Transformers, NLP, Transformer Models, LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, Multimodal AI, RoBERTa, Deep Learning, Model Evaluation.
- Languages: Python, TypeScript, JavaScript, SQL, Java, C++.
- Systems & Infrastructure: FastAPI, Node.js, Express, AsyncIO, Redis, BullMQ, Prisma ORM, RESTful APIs, WebSockets, MongoDB, JWT Authentication.
- Cloud & Deployment: Docker, Azure Fundamentals, Linux Environments, GitHub Actions.

PROJECT EXPERIENCE

Sentia: Multimodal Emotion Drift Analytics Platform

| [GitHub Repository](#) | [Technical Report](#)

Tech: Python, PyTorch, Hugging Face, RoBERTa, FastAPI, WebSockets

- Fine-tuned a RoBERTa-base transformer on 58k GoEmotions samples using a **normalized 7-class schema**, achieving a **macro F1-score above 0.70**.
- Improved multi-turn classification accuracy over text-only baselines by engineering a confidence-weighted late-fusion architecture combining text embeddings with a facial expression CNN.
- Architected a **7-stage parallel inference pipeline** executing concurrent text and vocal feature extraction loops via a ThreadPoolExecutor to maintain **P95 latency below 4 seconds**.
- Tracked emotional drift across multi-turn user sessions using probabilistic distance metrics (TVD and Euclidean shifts) over **7-class emotional distributions** to model emotional drift patterns across long-horizon interactions.
- Designed evaluation workflows to analyze behavioral consistency, emotional drift, and interaction safety patterns across long-horizon human-AI conversations.
- Developed a clause-level text segmentation pipeline to detect intra-message emotional transitions in conversational inputs.

Bhumi: Smart Land Governance AI Engine

| [GitHub Repository](#) | [Technical Report](#)

Tech: Python, LangChain, FastAPI, Vector Databases

- Architected an end-to-end Retrieval-Augmented Generation (RAG) pipeline using LangChain to ingest, segment, and index unstructured legal land records.
- Optimized semantic retrieval using Hugging Face embeddings and vector databases to improve relevance of domain-specific land-record question answering.
- Engineered an asynchronous FastAPI backend to process concurrent document indexing and user queries.

Panopticon: Distributed Political-Economic Data Observability Engine

| [GitHub Repository](#) | [Technical Report](#)

Tech: TypeScript, Node.js, Express, Prisma, BullMQ

- Integrated a LangChain-powered LLM processing layer into asynchronous worker pipelines to automate entity extraction from unstructured news feeds.
- Engineered a distributed tracing backend using TypeScript and Prisma ORM to aggregate and map multi-node political and economic data models.
- Developed asynchronous data-ingestion workers via BullMQ to process 10k+ structured events/day across automated cron cycles and task queues.
- Designed event-driven diagnostic telemetry loops using WebSockets to broadcast low-latency system-health metrics to a central monitoring cockpit.

OPEN-SOURCE CONTRIBUTIONS & LEADERSHIP

Project Admin / Contributor | GirlScript Summer of Code (GSSOC) | 2025-2026

- Project Admin (2026): Reviewed and merged contributor pull requests, maintained repository code health, and structured contribution guidelines across AI/ML and backend repositories.

- Contributor (2025-2026): Implemented modular feature extensions, resolved Git branch conflicts, and refined processing scripts across open-source AI/ML repositories.

AWARDS & ACHIEVEMENTS

- National Finalist | Smart India Hackathon (SIH), Ministry of Education, Govt. of India | 2024
- 1st Place, Smart India Hackathon Internal Selection | GLA University | 2024
- National Phase 2 Selection | IBM Praambh Hack Challenge | 2025
 - Advanced through competitive national screening evaluating technical system architecture and machine learning pipeline scalability.